

Logical Networks*

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Abstract

This is the short abstract.

1 Introduction

Artificial Intelligence systems have had a significant impact on the modern world over the decades. Some of the many advancements across various fields include (however certainly not limited to) conversational agents, marketing, medicine, and insurance (Nössig et al., 2024—need cite). Historically many of these advances and their impact on shaping the modern world have occurred in the background, improving our systems and applications iteratively and often at the periphery of the thinking of the general public. With new advances in AI, in particular machine learning methods (ML) and generative AI (in particular Large Language Models or LLMs), these systems have come to the forefront of public consciousness in a way that is likely to remain for the foreseeable future.

While these AI systems have the potential for significant advancement, a number of contemporary models come with significant drawbacks. In particular, the necessary growth of AI data centers and their associated infrastructure needs have caused significant stress for many local populations [Need citation]. These issues are leading to worsening availability of resources for the general

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public, and lead directly to the potential of significant environmental and health impacts (Bush 2025–need cite).

For continued advancement to avoid outpacing our environmental needs, significant research is necessary to bring down their respective footprints. Artificial Neural Network (ANN) technologies have made many of these technological advances possible, however drawbacks concerning their efficiency are a continued cause for concern regarding overall sustainability [cite?].

Answer Set Programming (ASP) languages often live at a different end of the research spectrum. These languages are designed to solve NP-Complete and NP-Hard problems in a way that is fast and efficient. In this paper we conduct investigate how to merge these schools of thought by building, representing, and performing inference over ANNs, specifically McCulloch-Pitts Networks (MPNs) [cite?]. To this end we introduce C-MCPN: The *Clingo McCullouch-Pitts Network Editor*.

In this paper we will introduce basic techniques for modeling ANNs in the ASP language Clingo, and demonstrate how C-MCPN can be used to operate over ANN models.

2 Background

2.1 Artificial Neural Networks (ANNs)

2.1.1 Efficiency and Environmental Impacts

2.2 Answer Set Programming (ASP)

Symbolic AI has served as a foundation for many of the earliest AI models and advancements. This form of AI relies on manipulation of symbols to represent knowledge and perform inference over models for the purpose of reasoning. These types of AI are explicitly rule-based, meaning that their decision-making systems are much more deterministic (Bhuyan et al., 2024; Liang et al., 2025; Meli et al., 2024–need cite).

The Prolog language is a classical example of these sorts of declarative (logical) programming language [cite]. Its syntax and semantics are largely derived from quantified, predicate logic, and has had a profound impact on the presentation of logic programming that continues into modern systems. Similarly, Boolean Satisfiability (SAT) solvers have served as a lower-level, contemporary form of logical programming. Throughout the 1990s and into the 2000s, these systems have expanded massively in their ability to solve large problems efficiently.

Modern ASP systems combine a simple, yet powerful syntax that is inspired by Prolog, with the efficiency of SAT systems. This empowers the user to write

concise programs that can solve computationally-difficult problem instances with rather remarkable speed and efficiency.

2.2.1 Syntax and Semantics